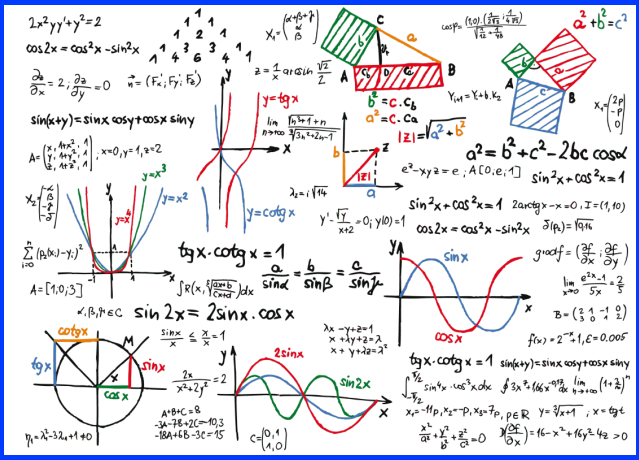
{EPITECH}

# 106BOMBYX_

< BOMBYX BOOMING BYLAW />



# 106BOMBYX

**binary name:** 106bombyx
**language:** everything working on ”the dump”
**compilation:** when necessary, via Makefile, including re, clean and fclean rules

- ✓ The totality of your source files, except all useless files (binary, temp files, objfiles,...), must be included in your delivery.
- ✓ All the bonus files (including a potential specific Makefile) should be in a directory named bonus.
- ✓ Error messages have to be written on the error output, and the program should then exit with the 84 error code (0 if there is no error).

In the 70's, chaos theory opened the way for a better understanding of the evolution of some animal species. Butterflies for instance. Let's look at... bombyx.

If a generation is crowded, chances are that the next generation will be crowded too, regarding the natural rules of reproduction. But resources may lack for this new generation, so it may not be able to develop. Therefore, the number of bombyx depends on those two contradictory factors, and its evolution is far from trivial.

Let's call $x_i$ the number of the $i^{th}$ generation of butterflies. Here is a model for the evolution of $x_i$:

- ✓ $x_i = n$ where $n$ is the number of first generation individuals ;
- ✓ $x_{i+1} = kx_i \frac{1000-x_i}{1000}$ for $i \geq 1$, $k$ being the *growth rate*, from 1 to 4.

In order to study this evolution, you are asked to plot two things:

- ✓ The curve representing the number of individuals in relation to the generation (varying from 1 to 100);
- ✓ A synthetic scheme summing all the results for a given $n$; it consists in plotting every value of $x_i$ (between two given bounds), in relation to $k$ ($k$ varying from 1 to 4 by 0.01 steps).

In both cases, your program shall print on the standard output the values to be entered into *gnuplot* to draw the graphs.

## Usage

Terminal

```
$> ./106bombyx -h
USAGE
    ./106bombyx n [k | i0 i1]

DESCRIPTION
    n       number of first generation individuals
    k       growth rate from 1 to 4
    i0      initial generation (included)
    i1      final generation (included)
```



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## Examples

Your program output has to be strictly identical to the one below.

Terminal

```
$> ./106bombyx 10 3.3 > data
$> head data
1 10.00
2 32.67
3 104.29
4 308.26
5 703.68
6 688.10
7 708.24
8 681.89
9 715.82
10 671.29
$> tail data
91 823.60
92 479.43
93 823.60
94 479.43
95 823.60
96 479.43
97 823.60
98 479.43
99 823.60
100 479.43
```

```
$> ./106bombyx 10 10000 10010 > data
$> head -n 30 data
1.00 0.10
1.00 0.10
1.00 0.10
1.00 0.10
1.00 0.10
1.00 0.10
1.00 0.10
1.00 0.10
1.00 0.10
1.00 0.10
1.00 0.10
1.01 9.90
1.01 9.90
1.01 9.90
1.01 9.90
1.01 9.90
1.01 9.90
1.01 9.90
1.01 9.90
1.01 9.90
1.01 9.90
1.01 9.90
1.02 19.61
1.02 19.61
1.02 19.61
1.02 19.61
1.02 19.61
1.02 19.61
1.02 19.61
1.02 19.61
$> head -n 1500 data | tail
2.35 574.47
2.35 574.47
2.35 574.47
2.35 574.47
2.35 574.47
2.35 574.47
2.36 576.27
2.36 576.27
2.36 576.27
2.36 576.27
```

Terminal

```
$> cat drawer.gnu
set terminal dumb
set nokey
plot "data"
```

Terminal

```
$> cat drawer.gnu | gnuplot

    1000 +-------------------------------------------------------------------+
         |          +           +          +          +           +    AAAAAA|
     900 |-+                                                       AAAAAAAA A-|
         |                                                   AAAAAAAAAAAAAAAA|
     800 |-+                                             AAAAA    AAAAAAAAAAA|
         |                                            AAA           AAAA AAA|
     700 |-+                                          A             AAAA AAA|
         |                                   AAAAAAAAA             AAAAAAA |
     600 |-+                            AAAAAAAA       AA           AAAAAAAAA|
         |                        AAAAAAA               AAA        AAAA AAAAA|
     500 |-+                  AAAAA                       AAAA   AAAAA AAAAAA|
         |                AAAA                                AAAA  AAAAA A A|
     400 |-+            AAAA                                   AAAAA  A AA-|
         |          AAA                                         AAAA  AA A |
     300 |-+      AAA                                             AAAAAA AA|
         |      AA                                                  AAA AAA|
     200 |-+  AAA                                                   AAAAAA|
         |  AAA                                                       AAAA|
     100 |-AA                                                          AAA|
         |AA        +           +          +          +           +        AA|
       0 +-------------------------------------------------------------------+
         1         1.5          2         2.5         3          3.5         4
```

v 3

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